CYBERSTER INTERNSHIP PROGRAM

# WEEK 12

# REPORT

## The Capstone Project – Full Attack Chain


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## Executive Summary

This report presents the activities completed during Week 12 of the Red Teaming Internship, focusing on planning, executing, documenting, and closing a complete simulated attack sequence within controlled laboratory environments. The objective was to combine the security testing concepts covered throughout the internship into a structured engagement that followed defined objectives, scope, evidence requirements, and cleanup procedures.

The week began with the Scoping and Planning Phase, where the authorized laboratory systems, network ranges, assessment objectives, Rules of Engagement, and success criteria were defined. A Pre-Engagement Document was prepared to establish the boundaries of the engagement and identify the systems and activities that were permitted during testing.

The second phase focused on the simulated attack chain, beginning with external reconnaissance and progressing through initial access, endpoint security assessment, privilege escalation, lateral movement, and Active Directory assessment. The final phases focused on cleanup and reporting — persistence mechanisms, temporary changes, created accounts, scheduled tasks, modified configurations, and other operational artifacts were reviewed and removed. Findings were organized into a final security report containing an executive summary, technical breakdown, risk ratings, MITRE ATT&CK mapping, and remediation recommendations.

## Introduction

A Red Team engagement involves more than identifying individual vulnerabilities. It requires structured planning, defined boundaries, evidence collection, controlled execution, and proper cleanup. A complete assessment follows an organized workflow in which the tester evaluates how weaknesses across different systems may contribute to a larger security risk.

During Week 12, the internship activities focused on combining previously learned reconnaissance, vulnerability assessment, initial access, privilege escalation, network pivoting, lateral movement, and Active Directory security concepts into a single simulated engagement. The activities were performed within controlled laboratory environments and followed predefined engagement boundaries.

This report documents the planning process, simulated attack progression, cleanup activities, collected evidence, and final reporting requirements. The objective was to demonstrate an organized approach to conducting and documenting a Red Team assessment from initial planning through final reporting.

## Objectives

The objectives of this week's practical activities were:

- To understand the importance of pre-engagement planning and defined Rules of Engagement.
- To define authorized targets, network ranges, objectives, and assessment boundaries.
- To establish measurable success criteria for the engagement.
- To combine previously learned security testing techniques into a structured attack workflow.
- To assess external attack surface and potential entry points within controlled environments.

- To evaluate the progression from initial access toward internal network and Active Directory resources.
- To maintain organized evidence throughout the engagement.
- To identify and document persistence mechanisms and system changes created during testing.
- To restore the laboratory environment to its original state after testing.
- To prepare a structured final security report.
- To prioritize findings using severity, impact, CVSS, and MITRE ATT&CK mapping.
- To develop appropriate remediation recommendations based on identified security risks.

## Lab Environment

The Week 12 activities were conducted using controlled and intentionally vulnerable laboratory environments. The environment included web applications, vulnerable virtual machines, Windows systems, and an Active Directory environment representing different stages of an enterprise security assessment.

| Component | Purpose |
|---|---|
| Hack This Site | External reconnaissance and web-based security testing. |
| Metasploitable 2 | Vulnerability assessment and initial access exercises. |
| Windows 7 Vulnerable VM | Windows security assessment. |
| OWASP Juice Shop | Web application security testing. |
| TryHackMe Attacktive Directory | Active Directory security assessment. |

## Tools Used

The following tools and technologies were used or studied during Week 12 practical activities:

| Tool | Purpose |
|---|---|
| Nmap | Network and service discovery |
| FFUF / Dirsearch | Web directory and endpoint discovery |
| WhatWeb | Web technology identification |
| Burp Suite | Web request analysis and testing |
| PowerShell | Windows and Active Directory administration and enumeration |
| PowerView | Active Directory enumeration |
| SharpHound | Active Directory relationship collection |
| BloodHound | Active Directory relationship analysis |
| LDAP Search | Directory information queries |
| ADExplorer | Graphical Active Directory analysis |
| Rubeus | Kerberos security assessment |
| Impacket Tools | Active Directory security assessment |
| Hashcat | Offline hash analysis |
| John the Ripper | Offline password analysis |

| Tool | Purpose |
| --- | --- |
| Kali Linux | Security testing platform |

## Methodology

The Week 12 engagement followed a structured four-phase methodology:

- Scoping and Planning – Define the authorized environment, objectives, Rules of Engagement, success criteria, and evidence requirements.
- Cyber Kill Chain Execution – Progress through reconnaissance, initial access, endpoint security assessment, privilege assessment, lateral movement, and Active Directory assessment.
- Post-Engagement Cleanup – Identify and remove persistence mechanisms, temporary accounts, modified configurations, and other assessment artifacts.
- Final Security Reporting – Document the attack progression, findings, evidence, risk ratings, MITRE ATT&CK mappings, and remediation recommendations.

This methodology provided a clear workflow from initial planning through final reporting and environment restoration.

## Task 1: Scoping & Planning Phase

### Objective

Define the scope, objectives, Rules of Engagement, success criteria, and evidence requirements before beginning technical security testing. The planning phase established clear boundaries for the engagement and ensured that testing remained limited to authorized laboratory systems.

### 7.1 Engagement Overview

The engagement was designed as a simulated Red Team assessment against a controlled corporate-style environment. The purpose was to evaluate the security of exposed services, internal systems, network segmentation, privilege management, and Active Directory resources.

| Item | Description |
|---|---|
| Assessment Type | Simulated Red Team Assessment |
| Environment | Controlled Home Lab |
| Primary Objective | Identify and assess security weaknesses |
| Assessment Scope | Authorized laboratory systems and networks |
| Assessment Approach | End-to-end security assessment |
| Evidence | Screenshots, commands, findings, timeline, and mappings |

### 7.2 Scope Definition

The scope definition identified the systems and networks authorized for testing. Only systems belonging to the authorized home laboratory environment were considered in scope. Personal devices, the hypervisor host, third-party systems, and other unauthorized resources were excluded from testing.

| Hostname | Role | IP Address |
|---|---|---|
| WEB01 | Public Web Server | 10.10.10.20 |
| FILE01 | File Server | 10.10.20.15 |
| DC01 | Domain Controller | 10.10.30.10 |

### 7.3 In-Scope Networks

The authorized network ranges were documented before technical activities began.

| Network | Description |
|---|---|
| 10.10.10.0/24 | DMZ Network |
| 10.10.20.0/24 | Internal User Network |
| 10.10.30.0/24 | Active Directory Network |

### 7.4 Out-of-Scope Assets

The following systems and activities were excluded from the engagement:

- Hypervisor host machine

- Personal devices and instructor systems
- Internet-facing third-party services
- Denial-of-Service attacks
- Destructive malware, data destruction, and ransomware simulation
- Persistent access after completion

### 7.5 Rules of Engagement

The Rules of Engagement defined the activities permitted and prohibited during the assessment.

**Allowed Activities**

- Enumeration
- Web application testing
- Controlled password security testing
- Privilege assessment and lateral movement assessment
- Active Directory Enumeration

**Prohibited Activities**

- Data destruction
- Ransomware simulation and destructive malware
- Denial-of-Service attacks
- Persistent access after completion

### 7.6 Engagement Objectives (Flags)

The engagement objectives were defined as measurable flags representing important stages of the simulated assessment.

| Flag ID | Objective |
|---|---|
| FLAG-01 | Identify exposed services |
| FLAG-02 | Gain initial access |
| FLAG-03 | Obtain local administrator privileges |
| FLAG-04 | Access sensitive file share |
| FLAG-05 | Assess Domain Administrator access |

### 7.7 Success Criteria

| Stage | Criteria |
|---|---|
| Reconnaissance | Identify exposed services and active hosts |
| Initial Access | Obtain a foothold on an authorized target |
| Privilege Escalation | Demonstrate administrative privilege escalation in the lab |
| Credential Access | Identify authentication-related security weaknesses |
| Lateral Movement | Demonstrate access to an additional authorized internal system |
| Active Directory Enumeration | Identify domain users, groups, shares, and relationships |
| Sensitive Data Access | Access authorized restricted laboratory resources |
| Domain-Level Assessment | Assess domain-level administrative security |

| Stage | Criteria |
| --- | --- |
| Cleanup | Remove assessment artifacts and restore the environment |

**Task 1 Summary:** The Scoping and Planning Phase established the foundation for the Week 12 engagement. The authorized systems, network ranges, objectives, Rules of Engagement, success criteria, and evidence requirements were defined before technical activities began. This ensured that the assessment remained structured, measurable, and limited to the designated laboratory environment.

# Task 2: Execution of the Cyber Kill Chain

## Objective

Combine the security testing concepts learned during previous internship activities into a continuous simulated attack workflow. The sequence progressed from external reconnaissance through initial access, endpoint security assessment, privilege assessment, internal movement, and Active Directory assessment.

## 8.1 External Reconnaissance

### Objective

Identify exposed assets, map the external attack surface, and identify potential entry points within the authorized environment.

### Activities Performed

- Performed passive reconnaissance.
- Identified externally exposed assets and enumerated available services.
- Identified technologies and frameworks.
- Reviewed exposed login points and documented potential entry points.
- Recorded relevant reconnaissance findings.

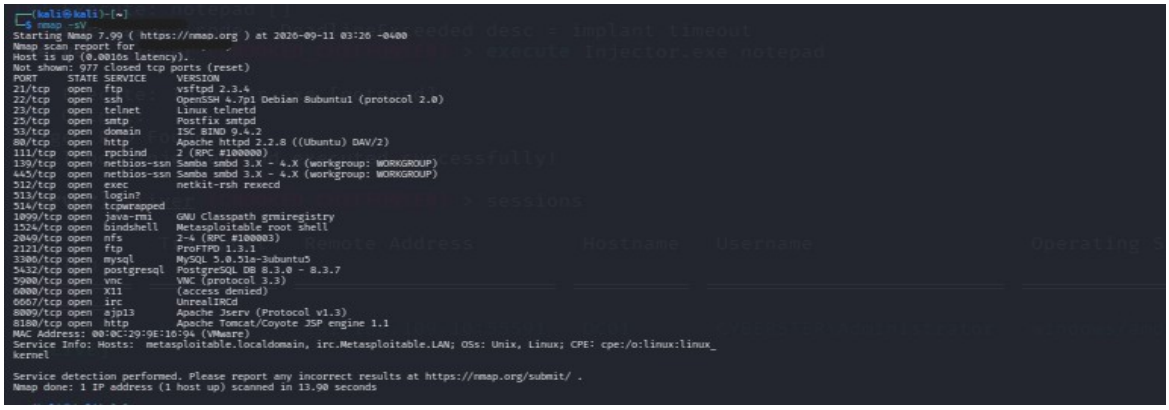
### Commands Used

```
nmap -sV <LAB-TARGET-IP>
whatweb http://127.0.0.1
dirsearch -u http://127.0.0.1
```

### Findings

The reconnaissance phase identified multiple exposed services and web technologies within the authorized laboratory environment. The Nmap scan of Metasploitable 2 identified services including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, MySQL, PostgreSQL, VNC, IRC, and Apache Tomcat. Web technology and directory enumeration provided additional information about accessible applications and potential entry points.

## Task 2 Screenshots



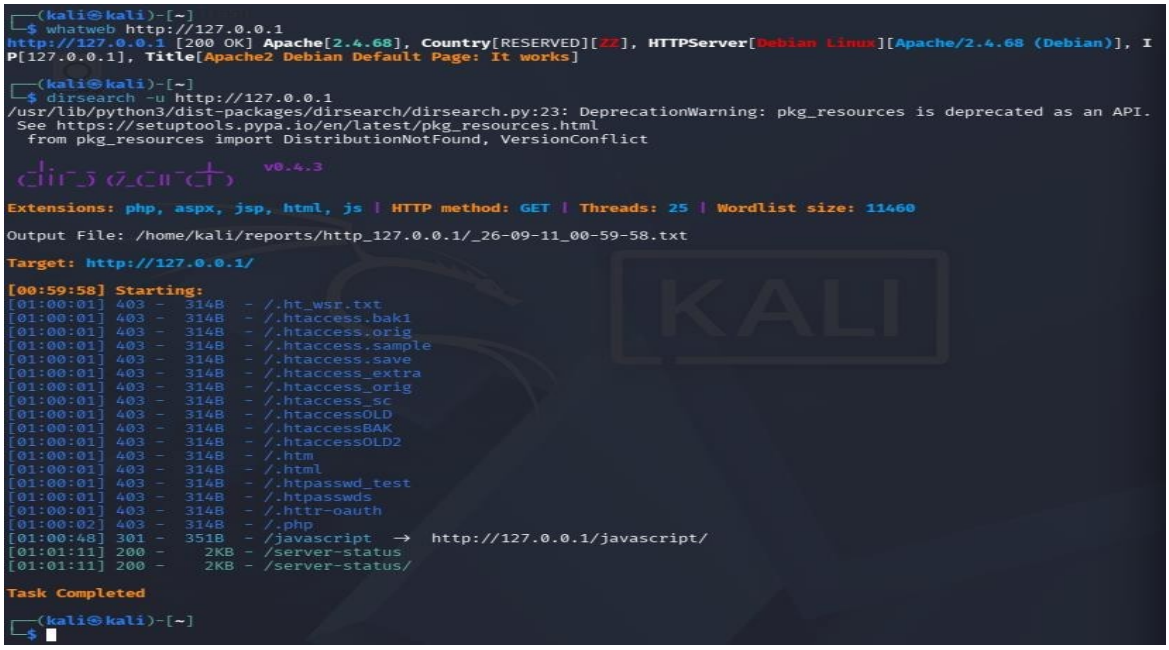



*Figure 8.1: External Reconnaissance and Service Enumeration Results*

## 8.2 Initial Access

### Objective

Assess whether vulnerabilities or authorized test credentials could provide an initial foothold within the laboratory environment.

### Activities Performed

- Performed web application enumeration.
- Reviewed application parameters and directories.
- Correlated identified services with known vulnerabilities.
- Validated security weaknesses within the controlled environment.
- Confirmed the resulting level of access where applicable.

### Findings

The initial access stage was reviewed against the authorized laboratory targets to determine whether the identified weaknesses could result in a valid foothold. Where access was established, the target identity and system information were verified and documented as supporting evidence. The results were used to determine the starting point for subsequent security assessment activities.

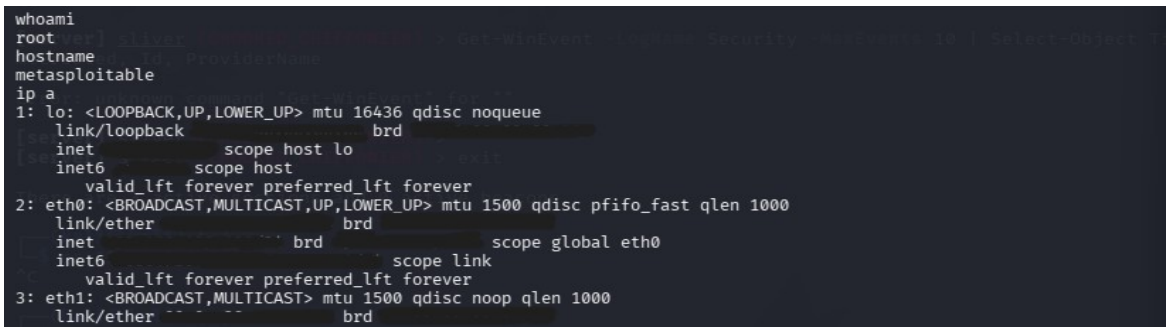



*Figure 8.2: Initial Access Verification*

## 8.3 Endpoint Security Assessment

### Objective

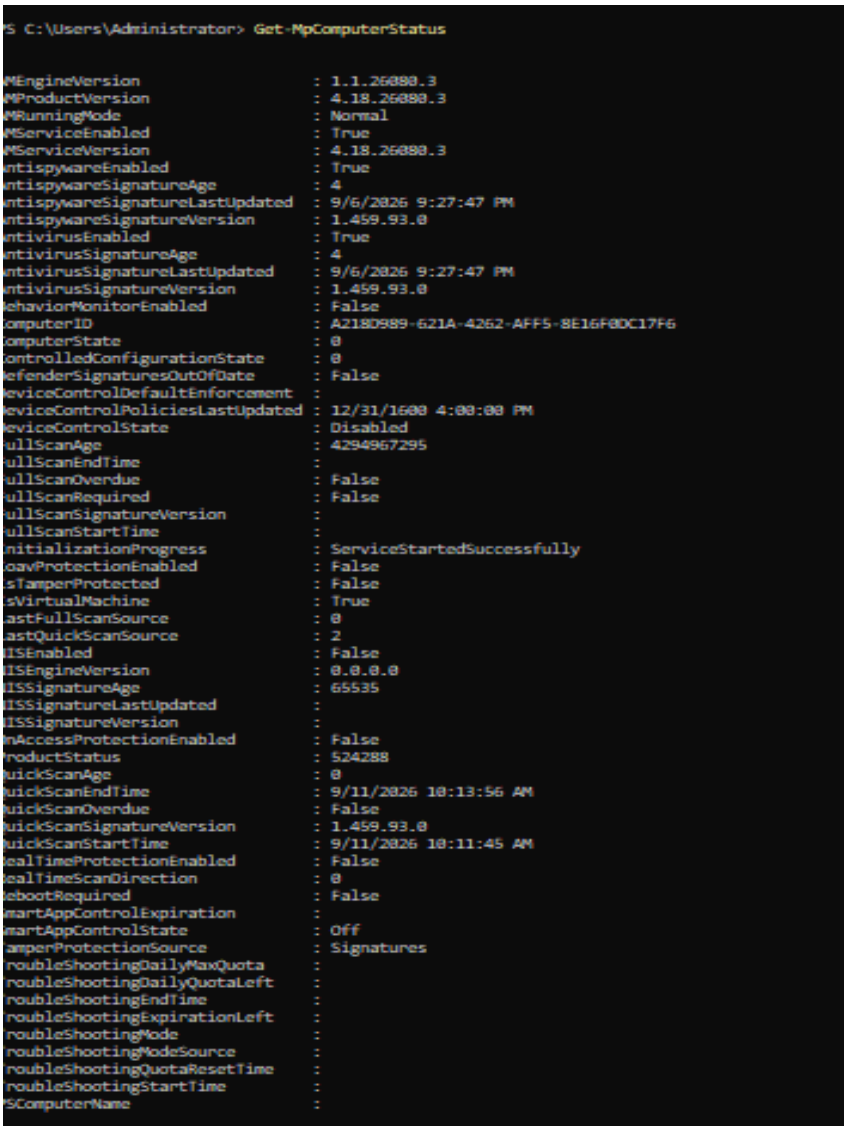
Assess the effectiveness of endpoint security controls within the laboratory environment.

### Activities Performed

- Reviewed endpoint protection configuration.
- Checked Microsoft Defender protection status.
- Reviewed available endpoint security controls.
- Documented the observed protection state.
- Evaluated the configured endpoint security posture.

### Findings

The Windows Endpoint security configuration was reviewed using Microsoft Defender status information. The collected output provided evidence of the configured endpoint protection state and supported evaluation of the defensive posture of the Windows system. The assessment documented the available protection status without making additional detection or blocking claims that were not directly observed.



```
S C:\Users\Administrator> Get-MpComputerStatus


MEngineVersion                  : 1.1.26080.3
MProductVersion                 : 4.18.26080.3
MRunningMode                    : Normal
MServiceEnabled                 : True
MServiceVersion                 : 4.18.26080.3
ntispywareEnabled               : True
ntispywareSignatureAge          : 4
ntispywareSignatureLastUpdated  : 9/6/2026 9:27:47 PM
ntispywareSignatureVersion      : 1.459.93.0
ntivirusEnabled                 : True
ntivirusSignatureAge            : 4
ntivirusSignatureLastUpdated    : 9/6/2026 9:27:47 PM
ntivirusSignatureVersion        : 1.459.93.0
ehaviorMonitorEnabled           : False
omputerID                       : A218D989-621A-4262-AFF5-8E16F0DC17F6
omputerState                    : 0
ontrolledConfigurationState     : 0
efenderSignaturesOutOfDate      : False
eviceControlDefaultEnforcement  :
eviceControlPoliciesLastUpdated : 12/31/1600 4:00:00 PM
eviceControlState               : Disabled
ullScanAge                      : 4294967295
ullScanEndTime                  :
ullScanOverdue                  : False
ullScanRequired                 : False
ullScanSignatureVersion         :
ullScanStartTime                :
nitializationProgress           : ServiceStartedSuccessfully
oavProtectionEnabled            : False
sTamperProtected                : False
sVirtualMachine                 : True
astFullScanSource               : 0
astQuickScanSource              : 2
ISEnabled                       : False
ISEngineVersion                 : 0.0.0.0
ISSignatureAge                  : 65535
ISSignatureLastUpdated          :
ISSignatureVersion              :
nAccessProtectionEnabled        : False
roductStatus                    : 524288
uickScanAge                     : 0
uickScanEndTime                 : 9/11/2026 10:13:56 AM
uickScanOverdue                 : False
uickScanSignatureVersion        : 1.459.93.0
uickScanStartTime               : 9/11/2026 10:11:45 AM
ealTimeProtectionEnabled        : False
ealTimeScanDirection            : 0
ebootRequired                   : False
martAppControlExpiration        :
martAppControlState             : Off
amperProtectionSource           : Signatures
roubleShootingDailyMaxQuota     :
roubleShootingDailyQuotaLeft    :
roubleShootingEndTime           :
roubleShootingExpirationLeft    :
roubleShootingMode              :
roubleShootingModeSource        :
roubleShootingQuotaResetTime    :
roubleShootingStartTime         :
SComputerName                   :
```

*Figure 8.3: Endpoint Security Control Response*

## 8.4 Privilege Escalation & Persistence Assessment

### Objective

Assess available privilege escalation paths and review persistence-related security risks on compromised laboratory systems.

### Activities Performed

- Reviewed possible privilege escalation paths.
- Identified weak permissions and configuration weaknesses.
- Assessed administrative privilege boundaries.
- Reviewed access to restricted resources.
- Documented persistence mechanisms used for the laboratory exercise.
- Recorded all system modifications for later cleanup.

### Findings

The assessment demonstrated the importance of properly configured permissions and administrative controls. Any persistence-related changes introduced during the laboratory exercise were recorded so they could be removed during the post-engagement cleanup phase.

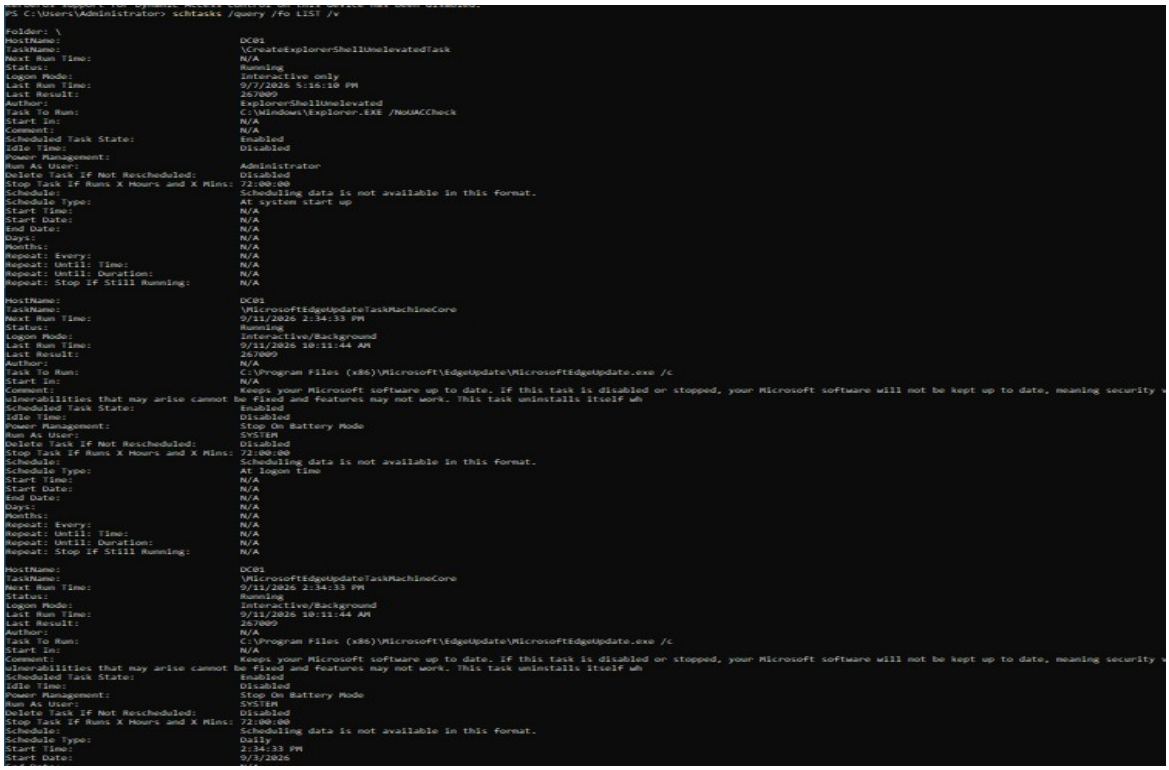



*Figure 8.4: Privilege Level Verification*

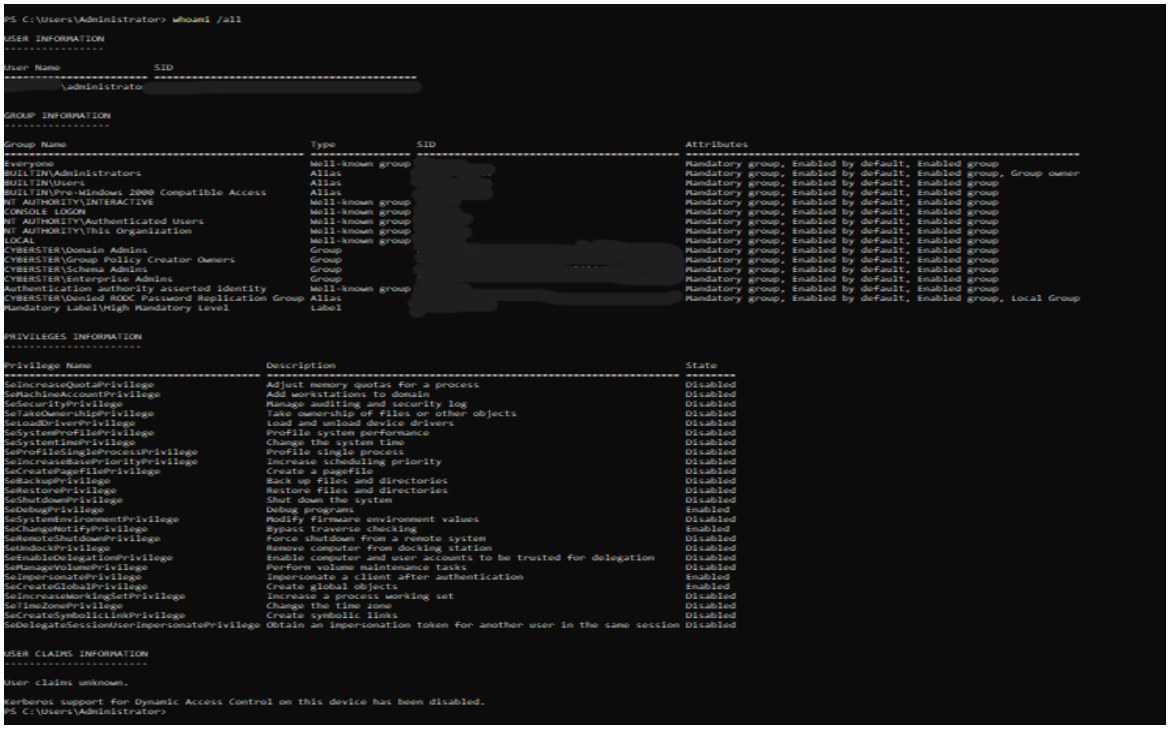



*Figure 8.5: Persistence Mechanism Documentation*

## 8.5 Pivoting & Lateral Movement

### Objective

Assess movement between authorized internal systems and understand how a compromised host could provide access to additional network resources.

### Activities Performed

- Performed internal network discovery.
- Reviewed accessible internal systems and assessed network segmentation.
- Used controlled tunneling techniques where applicable.
- Reviewed access relationships between systems.
- Documented reachable internal network segments and potential movement paths.

### Findings

The network routing information provided visibility into the interfaces and routes available from the assessed Windows system. This supported the review of reachable internal network segments and potential paths for internal movement. The collected evidence documented network visibility without claiming successful lateral movement where direct access to an additional system was not demonstrated.

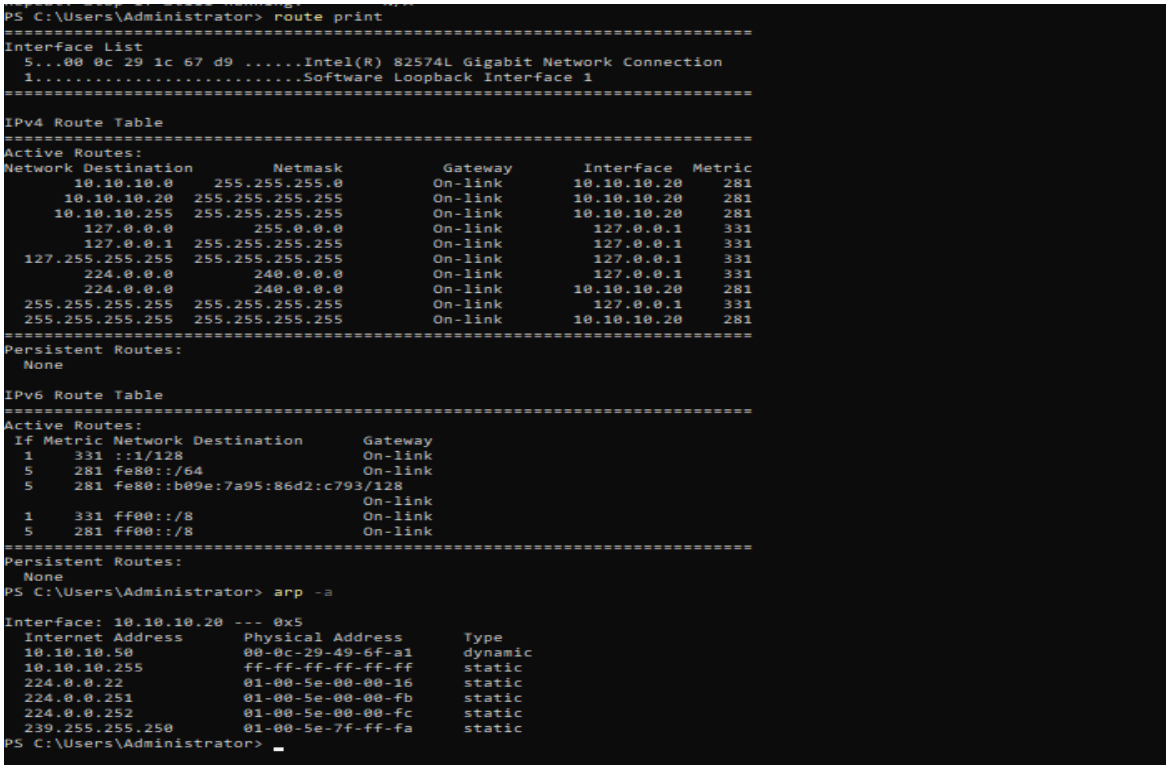



*Figure 8.6: Internal Network Routing and Pivoting Assessment*

## 8.6 Active Directory & Domain-Level Assessment

### Objective

Assess the Active Directory environment and identify relationships, permissions, and security weaknesses that could contribute to domain-level compromise.

### Activities Performed

- Enumerated domain users and groups.
- Reviewed domain computers and examined administrative groups.
- Analyzed Active Directory relationships and permissions.
- Reviewed information related to high-value domain resources.
- Documented relevant attack paths and security weaknesses.

### Commands Used

```
net group "Domain Admins" /domain
.\SharpHound.exe -c All
```

### Findings

The Active Directory assessment provided information about domain administrative groups and Active Directory relationships. Domain group enumeration provided visibility into administrative group membership, while SharpHound collection provided Active Directory information for reviewing relationships and potential access paths within the domain environment.

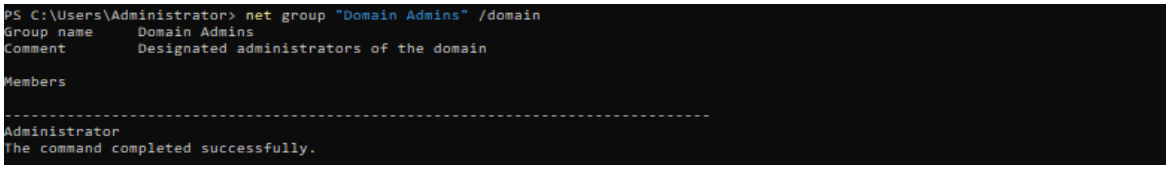



*Figure 8.7: Domain Administrative Group Enumeration*

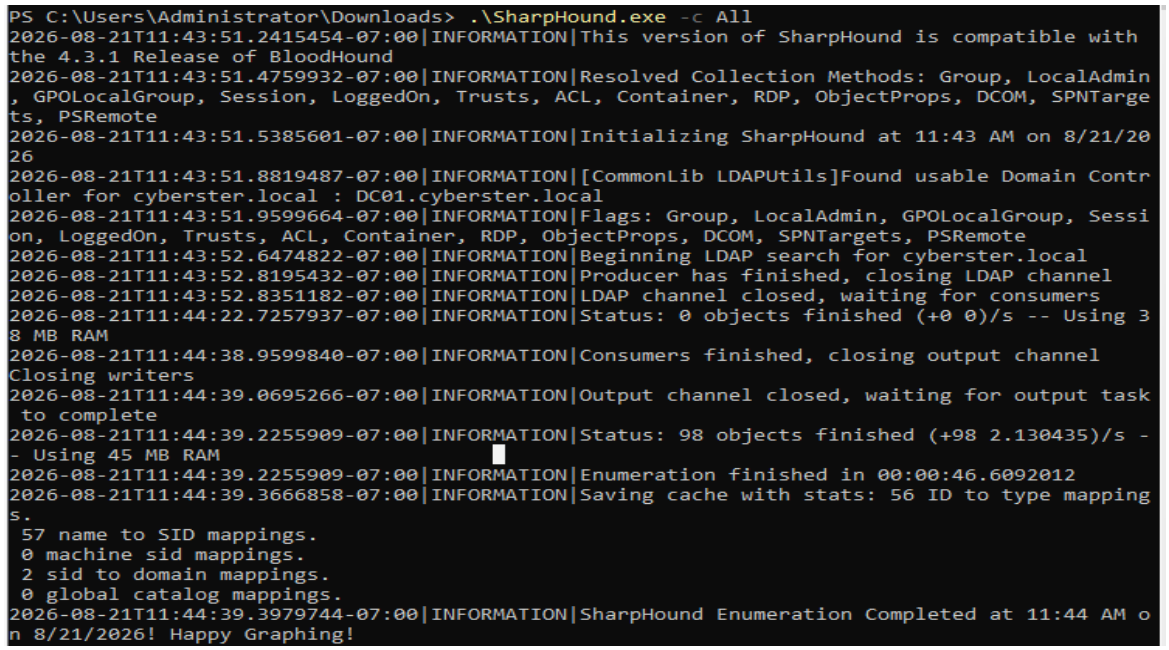



*Figure 8.8: SharpHound Active Directory Enumeration and Data Collection*

**Task 2 Summary:** The Cyber Kill Chain phase connected multiple security assessment stages into a structured workflow. The activities progressed from external reconnaissance and initial access toward endpoint security assessment, privilege assessment, internal network access, and Active Directory analysis. The collected evidence provided a documented record of the assessment progression within the authorized laboratory environment.

## Task 3: Post-Engagement Cleanup

### Objective

Identify all changes introduced during the engagement, remove unauthorized or temporary artifacts, and verify that the laboratory environment was restored to its original state.

### 9.1 Identification of Persistence Mechanisms

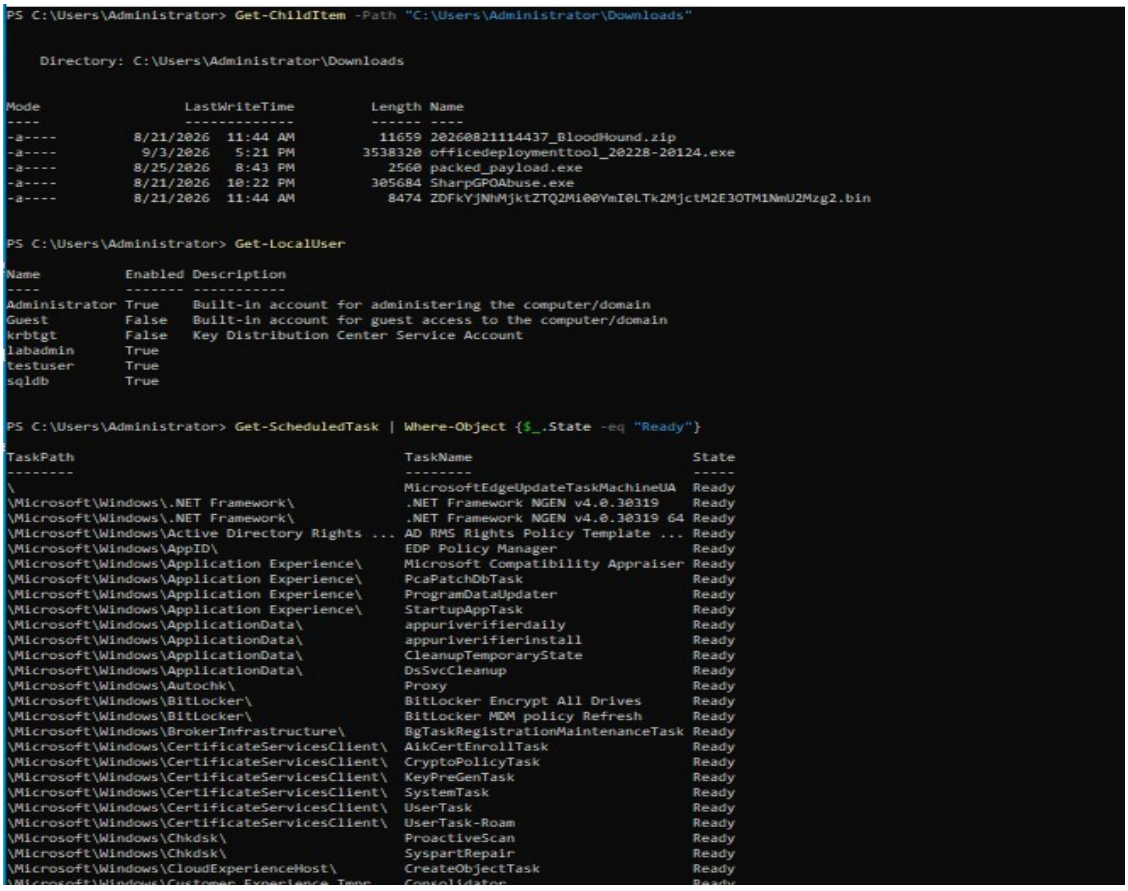
All persistence mechanisms and system modifications created during the assessment were reviewed. The review included:

- Temporary user accounts
- Scheduled tasks and services
- Startup entries and registry modifications
- Persistence scripts and uploaded tools or payloads
- Modified permissions and other temporary system changes

A record of each modification was maintained so that the affected system could be restored correctly.

**Findings**

The cleanup review identified the assessment-related changes and artifacts that required removal or restoration.



```
PS C:\Users\Administrator> Get-ChildItem -Path "C:\Users\Administrator\Downloads"


    Directory: C:\Users\Administrator\Downloads


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         8/21/2026  11:44 AM          11659 20260821114437_BloodHound.zip
-a----          9/3/2026   5:21 PM        3538320 officedeploymenttool_20228-20124.exe
-a----         8/25/2026   8:43 PM           2560 packed_payload.exe
-a----         8/21/2026  10:22 PM         305684 SharpGPOAbuse.exe
-a----         8/21/2026  11:44 AM           8474 ZDFkYjNhMjktZTQ2Mi00YmI0LTk2MjctM2E3OTM1NmU2Mzg2.bin


PS C:\Users\Administrator> Get-LocalUser

Name          Enabled Description
----          ------- -----------
Administrator True    Built-in account for administering the computer/domain
Guest         False   Built-in account for guest access to the computer/domain
krbtgt        False   Key Distribution Center Service Account
labadmin      True
testuser      True
sqldb         True


PS C:\Users\Administrator> Get-ScheduledTask | Where-Object {$_.State -eq "Ready"}

TaskPath                                    TaskName                          State
--------                                    --------                          -----
\                                           MicrosoftEdgeUpdateTaskMachineUA  Ready
\Microsoft\Windows\.NET Framework\          .NET Framework NGEN v4.0.30319    Ready
\Microsoft\Windows\.NET Framework\          .NET Framework NGEN v4.0.30319 64 Ready
\Microsoft\Windows\Active Directory Rights ... AD RMS Rights Policy Template ... Ready
\Microsoft\Windows\AppID\                   EDP Policy Manager                Ready
\Microsoft\Windows\Application Experience\  Microsoft Compatibility Appraiser Ready
\Microsoft\Windows\Application Experience\  PcaPatchDbTask                    Ready
\Microsoft\Windows\Application Experience\  ProgramDataUpdater                Ready
\Microsoft\Windows\Application Experience\  StartupAppTask                    Ready
\Microsoft\Windows\ApplicationData\         appuriverifierdaily               Ready
\Microsoft\Windows\ApplicationData\         appuriverifierinstall             Ready
\Microsoft\Windows\ApplicationData\         CleanupTemporaryState             Ready
\Microsoft\Windows\ApplicationData\         DsSvcCleanup                      Ready
\Microsoft\Windows\Autochk\                 Proxy                             Ready
\Microsoft\Windows\BitLocker\               BitLocker Encrypt All Drives      Ready
\Microsoft\Windows\BitLocker\               BitLocker MDM policy Refresh      Ready
\Microsoft\Windows\BrokerInfrastructure\    BgTaskRegistrationMaintenanceTask Ready
\Microsoft\Windows\CertificateServicesClient\ AikCertEnrollTask               Ready
\Microsoft\Windows\CertificateServicesClient\ CryptoPolicyTask                Ready
\Microsoft\Windows\CertificateServicesClient\ KeyPreGenTask                   Ready
\Microsoft\Windows\CertificateServicesClient\ SystemTask                      Ready
\Microsoft\Windows\CertificateServicesClient\ UserTask                        Ready
\Microsoft\Windows\CertificateServicesClient\ UserTask-Roam                   Ready
\Microsoft\Windows\Chkdsk\                  ProactiveScan                     Ready
\Microsoft\Windows\Chkdsk\                  SyspartRepair                     Ready
\Microsoft\Windows\CloudExperienceHost\     CreateObjectTask                  Ready
\Microsoft\Windows\Customer Experience Impro Consolidator                     Ready
```

*Figure 9.1: Identification of Assessment Artifacts*

### 9.2 Reverting Assessment Changes

The changes introduced during testing were reverted according to the recorded activity log. The cleanup process included:

- Removing temporary accounts.
- Removing temporary scheduled tasks.
- Reverting configuration changes.
- Removing uploaded assessment files and temporary scripts.
- Restoring modified permissions.
- Removing temporary services or startup entries where applicable.

**Findings**

The documented assessment changes were reviewed and reverted to reduce the possibility of leaving unauthorized access methods or temporary assessment artifacts on the laboratory systems.




*Figure 9.2: Cleanup and Restoration Process*

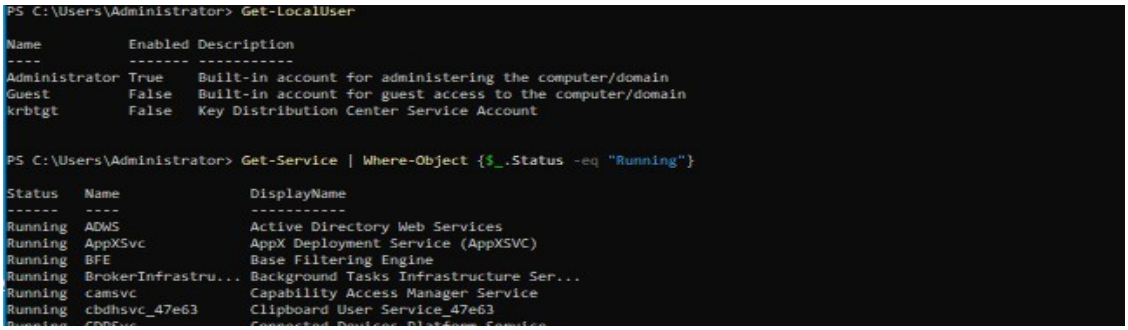
## 9.3 Verification of Original State

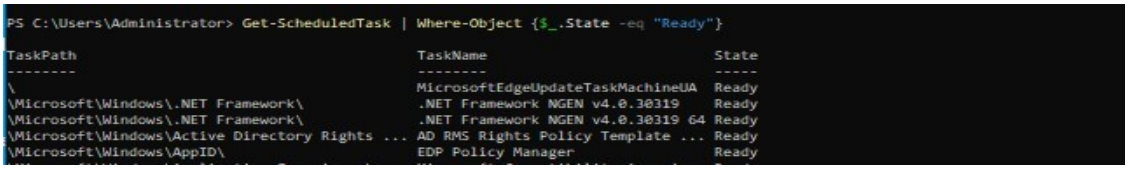
After completing the cleanup process, the systems were reviewed to confirm that the environment had returned to its expected state. The verification included:

- Reviewing user accounts and group memberships.
- Checking services and running processes.
- Reviewing scheduled tasks and startup locations.
- Reviewing modified configurations.
- Confirming network accessibility and verifying normal system operation.
- Performing a final system check after cleanup.

**Findings**

The final verification reviewed local user accounts, scheduled tasks, services, running processes, and network configuration to confirm that assessment-related changes had been removed or reverted where applicable.



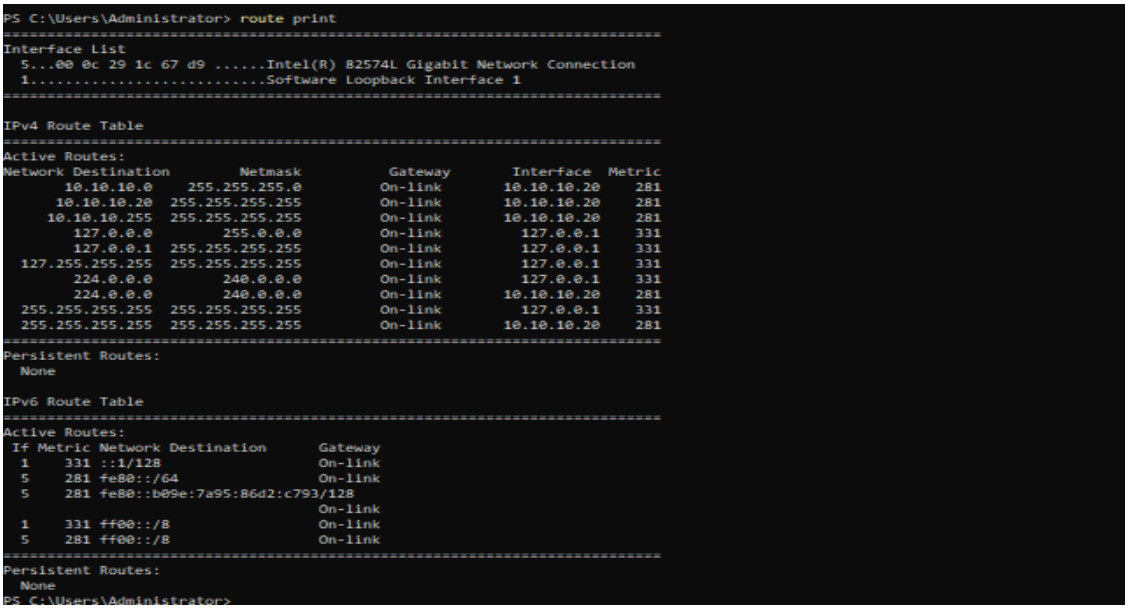



*Figure 9.3: Final System State Verification*

**Task 3 Summary:** The Post-Engagement Cleanup phase demonstrated the importance of responsible assessment closure. All identified modifications and assessment artifacts were reviewed, removed, or reverted where applicable. Final verification was performed to confirm that the laboratory environment remained stable and that no unnecessary assessment artifacts were left behind.

# Task 4: Final Security Report

## Objective

Organize the collected evidence and security findings into a structured final security report. The report summarizes the assessment progression, technical evidence, identified security risks, MITRE ATT&CK mappings, and recommended security improvements.

## 10.1 Executive Summary

The Week 12 engagement demonstrated how individual security testing activities can be combined into a structured Red Team workflow within a controlled laboratory environment. The engagement progressed from scope definition and reconnaissance through initial access assessment, endpoint security review, privilege assessment, internal network visibility, and Active Directory enumeration.

The collected evidence showed that exposed network services, administrative privileges, scheduled tasks, internal network visibility, and Active Directory relationships are important areas to review when evaluating the overall security posture of an environment. The engagement also demonstrated the importance of maintaining evidence throughout the testing process and removing assessment-related changes before completion.

Overall, the findings indicate that security should be evaluated across multiple layers rather than focusing on individual vulnerabilities only. Appropriate service exposure, least-privilege controls, network segmentation, Active Directory security, endpoint protection, and continuous monitoring can reduce the potential impact of a successful compromise.

## 10.2 Technical Breakdown

The technical breakdown summarizes the assessment activities in chronological order using the evidence documented in Tasks 2 and 3.

### Reconnaissance and Initial Access

The engagement began with external reconnaissance and service enumeration to identify exposed services, web technologies, directories, and potential entry points. The identified information was then used to assess the possibility of obtaining initial access within the authorized laboratory environment.

### Endpoint Security Assessment

After the initial access stage, the Windows Endpoint security configuration was reviewed to understand the available protection status and defensive controls.

### Privilege and Persistence Assessment

Privilege information and persistence-related locations were reviewed to understand administrative boundaries and identify system changes that required cleanup.

### Pivoting and Internal Network Assessment

Network interfaces and routing information were reviewed to understand the internal network visibility available from the assessed system. The collected evidence supported the evaluation of accessible network segments without claiming successful lateral movement where direct evidence was not available.

### Active Directory Assessment

The Active Directory environment was reviewed to identify administrative group membership and collect information about domain relationships and potential access paths. The collected information supported the assessment of domain-level security controls.

**Post-Engagement Cleanup**

After completing the assessment activities, temporary accounts, scheduled tasks, uploaded files, configuration changes, and other assessment artifacts were reviewed and removed where applicable. Final verification was performed to confirm that the laboratory environment returned to its expected state.

## 10.3 Risk Rating

The identified findings were categorized according to severity, exploitability, impact, and overall organizational risk. CVSS was used to represent vulnerability severity, while MITRE ATT&CK mappings were used to associate security activities with relevant adversary techniques.

**Risk Assessment Criteria**

| Finding | Affected Target | CVSS | Risk Level | MITRE ATT&CK | Description |
|---|---|---|---|---|---|
| Exposed Network Services | Metasploitable 2 | N/A* | High | T1046 – Network Service Scanning | Multiple services identified during enumeration. |
| Privilege & Admin Control | Windows Lab VM | N/A* | Medium | T1069 – Permission Groups Discovery | User, group, and privilege information reviewed. |
| Scheduled Task Exposure | Windows Lab VM | N/A* | Medium | T1053.005 – Scheduled Task/Job | Scheduled tasks reviewed for persistence. |
| Internal Network Visibility | Windows Lab VM | N/A* | Medium | T1016 – System Network Config Discovery | Routes and interfaces reviewed for lateral paths. |
| AD Admin Group Exposure | AD Environment | N/A* | Medium | T1069.002 – Domain Groups | Domain admin group membership enumerated. |
| AD Relationship Enumeration | AD Environment | N/A* | Medium | T1482 – Domain Trust Discovery | AD relationships collected for access-path analysis. |

**CVSS was marked N/A where the documented evidence represented a security observation or assessment activity rather than a specific vulnerability with sufficient information for a valid CVSS score.*

### 10.4 Remediation Recommendations

Based on the identified security weaknesses, the following remediation priorities should be considered:

- Remove unnecessary externally exposed services.
- Apply secure configuration and patch management practices.
- Enforce strong authentication and password policies.
- Apply the principle of least privilege and review administrative permissions regularly.
- Improve internal network segmentation and monitor authentication and lateral movement activity.
- Review Active Directory permissions and relationships.
- Enforce appropriate endpoint security controls.
- Remove unnecessary persistence mechanisms and temporary accounts.
- Maintain centralized logging and security monitoring.
- Conduct periodic security assessments to identify newly introduced weaknesses.

These recommendations should be prioritized according to the actual severity and business impact of the findings identified during the engagement.

**Task 4 Summary:** The final reporting phase converted the collected technical evidence into a structured security assessment. The report was organized to provide both high-level risk information and detailed technical evidence. Findings were prioritized using severity and impact, while CVSS and MITRE ATT&CK mappings were used to provide a standardized representation of the identified risks and activities.

## Learning Outcomes

During Week 12, I developed a broader understanding of how individual cybersecurity techniques can be combined into a structured Red Team engagement. I learned the importance of defining scope, Rules of Engagement, objectives, success criteria, and evidence requirements before beginning technical activities.

The engagement also improved my understanding of how reconnaissance, initial access, privilege assessment, internal network access, and Active Directory security assessment can be connected within a controlled environment. In addition, I gained experience in documenting system changes, performing post-engagement cleanup, and verifying that the laboratory environment was restored correctly.

The final reporting phase strengthened my ability to organize technical findings into an understandable security report. I also developed a better understanding of risk prioritization, CVSS severity assessment, MITRE ATT&CK mapping, business impact, and remediation planning.

## Conclusion

The Week 12 activities provided a structured understanding of the complete Red Team engagement lifecycle, beginning with scope definition and planning and progressing through simulated attack activities, cleanup, and final reporting. The engagement demonstrated the importance of connecting individual security testing techniques into an organized workflow while maintaining clear boundaries and evidence throughout the process.

The final stages emphasized that a professional security assessment does not end after identifying vulnerabilities. Proper cleanup, verification, documentation, risk prioritization, and remediation recommendations are equally important parts of the engagement. Overall, Week 12 strengthened my ability to plan, document, assess, and report security activities within a controlled laboratory environment.